

Isaac Mubiru

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PROFESSIONAL SUMMARY

Node.js engineer with 5+ years building **distributed data systems** in production. Built bidirectional replication between two live authoritative stores — concurrent writes on both sides, ordered Pub/Sub message processing, conflict handling — migrating 20+ applications with zero downtime and cutting ~\$5,000/month in infrastructure cost. Architected event-driven microservices on Apache Kafka, NATS Streaming and gRPC + Protobuf serving four tenants on shared infrastructure. Code at github.com/zac-09.

Deliberately moving into peer-to-peer: studying Kademlia DHTs, the Hypercore/hyperswarm stack, and distributed indexing — distributed systems and transport reliability are the foundation I already have, P2P and information retrieval are the specialisms I am building next. Bias for shipping: features on this CV carry revenue and cost numbers, not just architecture diagrams. 5+ years fully remote on US and Singapore teams from Uganda (UTC+3), async-first, direct CTO reporting line.

CORE COMPETENCIES

Node.js

Distributed Systems

Distributed Databases

Bidirectional Replication

Apache Kafka

NATS Streaming

gRPC + Protobuf

Transport Reliability

Concurrent Message Processing

Indexing & Query at Scale

Microservices

Async-First Remote

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote (from Uganda, UTC+3)

- Built **bidirectional replication between MongoDB and Firestore** in Node.js over Google Pub/Sub — two live authoritative stores accepting concurrent writes, with ordered message processing, conflict handling and a parallel write path enabling gradual cutover.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Saved the company **~\$5,000/month** by migrating all services off AWS; moved file storage from Amazon S3 to GCS via a Python pipeline.
- Shipped Express Shipping end to end, generating an additional **\$40 revenue per order**.
- Built a 3D visualization feature using video overlay and the ffmpeg library, increasing buyer conversion.

CodeBits

Jan 2020 – Jul 2021

Software Engineer · Lead of 4 · Uganda Remote

- Architected an **event-driven microservices backend** — Apache Kafka topics, NATS Streaming, and **gRPC + Protobuf** service contracts on Docker/Kubernetes — supporting four NGO tenants with independent service deploys.
- Built a USSD legal-aid service over an unreliable 2G carrier path: Node.js backend, gRPC inter-service communication, and a **retry-aware session state machine** that resumed dropped sessions instead of restarting them — transport reliability and multiplexing under constant packet loss.
- Led a team of 4 engineers: owned architecture, ran code review, and split delivery along service boundaries.
- Built a case management system for FIDA Uganda, a paralegal database for LASPNET, and a cross-platform React Native app with Firebase push notifications.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Built ad-hoc jobs querying **over 2 million Firestore records** with diff-detection to avoid full rescans, keeping real-time stock prices current from Morningstar APIs at sub-15-minute freshness.
- Led optimization of a PWA delivering real-time market prices from a Firebase backend.
- Extended backend APIs using Firebase Cloud Functions and Express.js.

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for **decentralized (DeFi) applications** using Web3 and Node.js — contract integration, transaction signing, off-chain indexing.
- Delivered React UIs from Figma designs under tight client timelines.

PROJECTS

- **Marketplace recommendation engine** — collaborative + content-based filtering in MongoDB for service-provider matching; relevance ranking and candidate retrieval over a live catalogue.
- **Healthcare WhatsApp assistant** — Node.js service integrating OpenAI and Google Vision for prescription OCR over the WhatsApp Business API.
- **Real-time Pub/Sub sync pipeline** — live MongoDB ↔ Firestore replication supporting parallel production traffic throughout a multi-app migration.

EDUCATION

B.Sc. Computer Engineering — [Makerere University](#)

Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, JavaScript, Firebase, GCP, Python, MongoDB, SQL, Linux, Git, Docker, Kubernetes, HTML/CSS, React

Distributed systems: Apache Kafka, NATS Streaming, gRPC, Protobuf, Google Pub/Sub, bidirectional replication, microservices

Intermediate: Go, Rust, AWS, React Native, Web3

Currently studying: Kademlia DHT, Hypercore / hyperswarm / hyperbee, distributed indexing, inverted indices and BM25 ranking

Languages: English (fluent, working language for 5+ years of remote work), Kiswahili